Hamilton Cycles in the Noncrossing Partition Refinement Graph

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Abstract

Let $\mathrm{NC}_R(n)$ be the cover graph of the refinement order on the noncrossing partitions of an n-element cyclically ordered set. Thus an edge merges two blocks, or reverses such a merge by splitting one block. We determine exactly when this graph has a Hamilton cycle. For odd $n \geqslant 3$, block-count parity is a bipartition and a signed Dyck-tree recurrence gives color-class difference $(-1)^{m+1}\mathrm{Cat}_m$ at $n = 2m+1$, so a Hamilton cycle is impossible. For every even $n \geqslant 4$, we give one uniform construction. A noncrossing partition is encoded bijectively by a noncrossing partial matching Q and a Boolean mask on the unmatched nonroot points. Each fixed Q is therefore an odd-dimensional hypercube and carries a cyclic reflected Gray code. Deleting the lexicographically first chord makes the matchings into a rooted tree. For every tree edge we exhibit an explicit refinement diamond; an injective port assignment makes all selected cube edges disjoint. Recursive square switching then joins all Boolean cycles into one Hamilton cycle. With the formal convention for singleton graphs, the complete classification for all natural n is

$$\mathrm{NC}_R(n) \text{ is Hamiltonian} \quad\iff\quad n \in \{0,1\} \text{ or } (n \text{ is even and } n \geqslant 4).$$

The construction, the odd obstruction, the small cases, and the final classification are formalized in Lean 4. The publication theorem has no project axiom; its axiom audit is exactly `[propext, Classical.choice, Quot.sound]`.

Mathematics Subject Classifications: 05A18, 05C45, 05C85

1 Introduction

Write $X_n = \{0, 1, \ldots, n-1\}$ with its cyclic order. A partition of X_n is *noncrossing* if there are no $a < b < c < d$ for which a, c belong to one block and b, d belong to another. The set $\mathrm{NC}(n)$ of all noncrossing partitions is the Kreweras lattice under refinement [4, 8, 9]. Its cover graph $\mathrm{NC}_R(n)$ has vertex set $\mathrm{NC}(n)$; two vertices are adjacent when one is obtained from the other by merging exactly two blocks, with the result still noncrossing.

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The adjacency here is not the element-transfer adjacency in the cyclic Gray code of Huemer, Hurtado, Noy, and Omaña-Pulido [2]. The general permutation-language results of Hartung–Hoang–Mütze–Williams and Hoang–Mütze [1, 3] provide Hamilton paths for many quotient graphs, but do not identify the refinement cover graph with such a quotient. The arbitrary-sublist refinement Gray-code problem considered by Merino, Namrata, and Williams [5] is also different from the spanning cycle problem for the full lattice. For broader background on combinatorial Gray codes and their Hamiltonian formulations, see Mütze's survey [6].

Petersen [7] constructed a symmetric Boolean decomposition of the noncrossing partition lattice using valley hopping. Our proof makes the corresponding coordinates direct: a Boolean block is indexed by a canonical noncrossing partial matching, and its Boolean coordinate is an explicit mask. This direct description is the one used in the machine proof.

Our main result is the following complete classification.

Theorem 1 (Complete classification). *Under the closed-spanning-walk convention used in the formalization, for every $n \in \mathbb{N}$,*

$$\mathrm{NC}_R(n) \text{ has a Hamilton cycle} \iff n = 0 \text{ or } n = 1 \text{ or } (n \text{ is even and } 4 \leqslant n).$$

Consequently, in the usual nondegenerate range $n \geqslant 3$, $\mathrm{NC}_R(n)$ has a Hamilton cycle if and only if n is even.

Sections 2 and 3–5 prove the two directions. Section 7 gives the exact correspondence between the displayed results and Lean declarations.

2 The odd-order obstruction

Let $b(\pi)$ denote the number of blocks of π. Every cover changes $b(\pi)$ by one, hence block-count parity colors the graph properly.

Lemma 2. *The graph $\mathrm{NC}_R(n)$ is bipartite, with color classes*

$$E_n = \{\pi : b(\pi) \text{ is even}\}, \qquad O_n = \{\pi : b(\pi) \text{ is odd}\}.$$

Proof. A cover relation merges two blocks or splits one block into two. Thus the two endpoints have block counts differing by exactly one, and therefore have opposite parity. □

Define the signed count

$$D_n := \sum_{\pi \in \mathrm{NC}(n)} (-1)^{b(\pi)} = |E_n| - |O_n|.$$

Under the standard Dyck-word, or equivalently plane-binary-tree, correspondence for noncrossing partitions, the number of blocks is the number of peaks. Decomposing a nonempty tree at its root yields the recurrence

$$D_{n+1} = -D_n + \sum_{k=0}^{n-1} D_{k+1} D_{n-1-k}, \qquad D_0 = 1, quad D_1 = -1. \tag{1}$$

Theorem 3 (Signed count)**.** *For $m \geqslant 1$,*

$$D_{2m} = 0, \qquad D_{2m+1} = (-1)^{m+1}\mathrm{Cat}_m,$$

where $\mathrm{Cat}_m = \frac{1}{m+1}\binom{2m}{m}$.

Proof. We use simultaneous strong induction on m. In (1) for D_{2m}, the two indices in every product sum to $2m-1$. Hence one index is positive and even, except for the boundary term $D_{2m-1}D_0$. Every nonboundary term vanishes by induction, and the boundary term cancels the initial $-D_{2m-1}$. Thus $D_{2m} = 0$.

For D_{2m+1}, all terms having a positive even index vanish. The remaining terms have indices $2j+1$ and $2(m-1-j)+1$. By induction their product is

$$(-1)^{j+1}\mathrm{Cat}_j(-1)^{m-j}\mathrm{Cat}_{m-1-j} = (-1)^{m+1}\mathrm{Cat}_j\mathrm{Cat}_{m-1-j}.$$

The initial term is zero because $D_{2m} = 0$. Summing and using the Catalan recurrence $\mathrm{Cat}_m = \sum_{j=0}^{m-1}\mathrm{Cat}_j\mathrm{Cat}_{m-1-j}$ proves the formula. The initial values $D_0 = 1$ and $D_1 = -1$ start the induction. □

Theorem 4 (Odd-order obstruction)**.** *If $n \geqslant 3$ is odd, then $\mathrm{NC}_R(n)$ has no Hamilton cycle.*

Proof. Write $n = 2m+1$. Since $n \geqslant 3$, $m \geqslant 1$, and Theorem 3 gives $|E_n| - |O_n| = (-1)^{m+1}\mathrm{Cat}_m \neq 0$. Every cycle in a bipartite graph uses equally many vertices from its two color classes. A spanning cycle would therefore force $|E_n| = |O_n|$, a contradiction. □

3 Canonical matching–mask coordinates

We now assume $n > 0$. A *canonical matching* is a set Q of ordered chords (a, b) with

$$0 < a < b < n,$$

such that distinct chords have disjoint endpoints and no two chords cross. The chords may be nested. Let $\mathrm{end}(Q)$ be their endpoint set and define the free points

$$F(Q) = X_n \setminus (\{0\} \cup \mathrm{end}(Q)).$$

For a chord endpoint x, let $e_Q(x)$ be the left endpoint of its chord. For $x \in F(Q)$, let $a_Q(x)$ be the left endpoint of the innermost chord whose open interval contains x, or 0 if there is no such chord.

For a mask $A \subseteq F(Q)$ define

$$\kappa_{Q,A}(x) = \begin{cases} e_Q(x), & x \in \mathrm{end}(Q), \\ a_Q(x), & x \in A, \\ x, & \text{otherwise}. \end{cases} \tag{2}$$

Let $P_Q(A)$ be the partition into the fibers of $\kappa_{Q,A}$.

Lemma 5 (Key-fiber geometry)**.** *If two distinct points have the same decoded key, then that key is either* 0 *or the left endpoint of a chord of Q. A fiber based at a chord (a,b) is contained in $[a,b]$. A point in that fiber cannot lie strictly inside a properly nested chord with a different left endpoint, and a point in the root fiber cannot lie strictly inside any chord of Q. Consequently, two distinct decoded fibers cannot cross.*

Proof. The first assertion follows directly from (2). A point can change its own key only by being a chord endpoint or by being selected in the mask. In the first case the common key is that chord's left endpoint. In the second it is the left endpoint of the innermost containing chord, or 0.

Fix $(a,b) \in Q$. Its endpoints decode to a. If another point x also decodes to a, then either x is one of these endpoints or x is selected and (a,b) is its innermost containing chord; in either case $a \leqslant x \leqslant b$. If (c,d) is properly nested in (a,b) and $c < x < d$, then (c,d) is a strictly inner containing chord, so the anchor of a selected x is at least $c > a$; hence x cannot decode to a. The same argument with no outer chord shows that a root-fiber point cannot lie inside any chord.

Suppose fibers based at r and s crossed, with $x_1 < y_1 < x_2 < y_2$ in the two fibers. If one key is 0, the other fiber's chord has both endpoints between two root-fiber points, contradicting the root assertion. Otherwise the two base chords are disjoint or nested because Q is noncrossing. The four alternating points force one fiber point into a properly nested chord with a different base, contradicting the preceding paragraph. $\square$

Theorem 6 (Boolean decomposition)**.** *For every $n > 0$, the map*

$$(Q, A) \longmapsto P_Q(A), \qquad Q \textit{ canonical}, \quad A \subseteq F(Q),$$

is a bijection onto $\mathrm{NC}(n)$*. Moreover, if $q \in F(Q) \setminus A$, then $P_Q(A \cup \{q\})$ covers $P_Q(A)$ in the refinement lattice.*

Proof. First note that $\kappa_{Q,A}(x) \leqslant x$. By Lemma 5, four alternating points cannot lie in two different fibers. Thus $P_Q(A)$ is noncrossing.

Conversely, for $\pi \in \mathrm{NC}(n)$, take from every nonsingleton block not based at 0 the chord joining its minimum and maximum. Distinct blocks give endpoint-disjoint noncrossing chords, so this is a canonical matching $Q(\pi)$. Put in $A(\pi)$ exactly those remaining nonroot, nonendpoint points whose block is nontrivial. The minimum of the block containing such a point is precisely the left endpoint of the innermost containing chord, or 0. Therefore

$$\kappa_{Q(\pi),A(\pi)}(x) = \min(\text{the block of } x)$$

for every x, and hence $P_{Q(\pi)}(A(\pi)) = \pi$.

The same description recovers Q from the extrema of the nonroot nonsingleton fibers of $P_Q(A)$, and then recovers A by testing which free points have nonsingleton fibers. The decoder is therefore injective and surjective. Finally, inserting q into the mask changes only its key, from q to $a_Q(q)$; it merges the singleton block $\{q\}$ with the block based at $a_Q(q)$ and leaves every other block unchanged. This is one refinement cover. $\square$

The dimension of the block indexed by Q is

$$d(Q) := |F(Q)| = n - 1 - 2|Q|. \tag{3}$$

If n is even, then $d(Q)$ is positive and odd.

4 Gray cycles, the block tree, and ports

Order $F(Q)$ increasingly and use the binary-reflected Gray list on its subsets. Adjacent masks differ by one free point, so Theorem 6 sends every Gray edge to a refinement edge.

Lemma 7 (Block Gray cycle)**.** *For even n, every block $\{P_Q(A) : A \subseteq F(Q)\}$ has a cyclic Gray listing that visits each of its vertices once. For $d(Q) \geqslant 3$ it is a simple cycle; for $d(Q) = 1$ it is the two directed occurrences of the unique edge, to be consumed by the tree switch below.*

Proof. The reflected recursion

$$G(x_0, x_1, \ldots) = G(x_1, \ldots), \mathrm{rev}(G(x_1, \ldots)) + x_0$$

lists every subset once, changes one coordinate at every step, and has a closing one-coordinate change. Equation (3) makes the dimension positive. Decoding gives the asserted refinement edges and the bijection gives distinctness and coverage. □

If $Q \neq \varnothing$, let $p(Q)$ be obtained by deleting the lexicographically first chord of Q.

Lemma 8 (Canonical block tree)**.** *The map p defines a rooted tree on canonical matchings, with root the empty matching. Each parent step decreases chord count by one and increases free dimension by two. Every canonical matching lies in the root subtree.*

Proof. Deleting a chord preserves endpoint disjointness and noncrossingness. It decreases $|Q|$ by one, so iteration terminates at the unique chordless matching. The parent is uniquely determined by the lexicographically first chord. Conversely, every nonroot matching has exactly that parent. Thus the directed parent relation is acyclic, connected to the empty matching, and has unique parent at every nonroot vertex; it is a rooted tree. Formula (3) gives the dimension change. □

Let C be a child of Q, obtained by inserting its first chord between the ith and jth free points of Q, $i < j$. The reflected Gray cycles admit the following explicit opposite edges. Indices in a mask refer to the increasing free-point order of the corresponding block.

(i,j)	parent edge in Q	child edge in C
$1 \leqslant i < j$	$\{i,j\}, \{0,i,j\}$	$\varnothing, \{0\}$
$i = 0,\ j \geqslant 2$	$\{0,j\}, \{0,1,j\}$	$\varnothing, \{0\}$
$(i,j) = (0,1)$	$\varnothing, \{d(Q)-1\}$	$\varnothing, \{d(C)-1\}$.

(4)

Lemma 9 (Explicit joining diamond). *For every parent–child pair Q, C, the two displayed Gray edges in the appropriate row of (4), together with two cross-block refinement edges, form a four-cycle in $\mathrm{NC}_R(n)$.*

Proof. Let (a, b) be the deleted chord. In the parent, a and b are free; in the child they are the endpoints of one matching chord. The decoder formula (2) shows that deleting or inserting this chord relabels exactly the two key classes based at a and at the common anchor of the two points. Therefore the child mask in the left column is related to the parent mask in the same row by one merge. Flipping the extra Gray coordinate shown in that row changes the same key in both blocks, so the second pair is again related by the same one-block merge. These are the two cross edges.

More explicitly, the cross covers in the first case are

$$P_C(\varnothing) \lessdot P_Q(\{i, j\}), \qquad P_C(\{0\}) \lessdot P_Q(\{0, i, j\});$$

in the second case they are

$$P_C(\varnothing) \lessdot P_Q(\{0, j\}), \qquad P_C(\{0\}) \lessdot P_Q(\{0, 1, j\});$$

and in the special case they are

$$P_Q(\varnothing) \lessdot P_C(\varnothing), \qquad P_Q(\{d(Q) - 1\}) \lessdot P_C(\{d(C) - 1\}).$$

Each relation changes exactly one key fiber and hence exactly one block. For $1 \leqslant i < j$ the horizontal parent edge flips coordinate 0; for $i = 0 < j$ with $j \geqslant 2$ it flips coordinate 1; and for $(i, j) = (0, 1)$ it is the closing Gray edge. The child edge is respectively its coordinate-0 edge or closing edge. Hence the two horizontal pairs are edges of their block cycles, and the four vertices form a literal refinement diamond. □

For bookkeeping, call a Gray edge's *port key* either (q, H), meaning "flip coordinate q above ambient lower mask H", or *closing*. The three rows of (4) give one incoming key in the parent and one outgoing key in the child.

Lemma 10 (Port injection). *For a fixed block Q, distinct children are assigned to distinct directed edge occurrences of the Gray cycle of Q. If Q is nonroot, its outgoing edge occurrence toward $p(Q)$ is not assigned to any child.*

Proof. A nonclosing Gray edge determines both its flipped coordinate and its lower mask, so it realizes at most one coordinate port key. The closing occurrence realizes only the closing key. In the first row of (4), the pair (i, j) is recovered from the lower mask $\{i, j\}$; in the second, j is recovered from $\{0, j\}$ and the flipped coordinate distinguishes the row; the third row is the unique closing key. Thus incoming keys of distinct children are distinct and each is realized by a directed Gray edge.

For a nonroot Q, its outgoing key is the child-side key of its own joining diamond. Comparing the three cases shows that this key differs from every incoming child key: a coordinate key differs in coordinate or lower mask, and a closing outgoing key occurs only in the special $(0, 1)$ case, where Q has no child whose first chord could precede that special chord. Therefore the chosen outgoing occurrence is not consumed by any child. □

5 Recursive square switching

We use the elementary square switch: if two disjoint cycles contain opposite edges ab and cd of a four-cycle $abdca$, deleting ab, cd and adding ac, bd produces one cycle on the union of their vertices. Applying switches sequentially is delicate because a parent cycle is modified by its earlier children. The following recursive formulation records exactly which edge occurrence remains available.

Lemma 11 (Block-tree assembly)**.** *For every canonical matching Q and even n, there is a cycle whose support is exactly the union of all Boolean blocks in the subtree rooted at Q. If Q is nonroot, the cycle retains the outgoing port prescribed by the diamond joining Q to $p(Q)$.*

Proof. Induct on $d(Q) = |F(Q)|$. Every child has free dimension $d(Q)-2$, so the induction is well founded. By induction each child subtree has one cycle and retains its outgoing edge. Traverse the directed edge occurrences of the Gray cycle of Q. At an occurrence assigned to a child, cut the parent edge and the retained child edge and insert the two cross edges of Lemma 9; regard the resulting child excursion as one path piece. At an unassigned occurrence, retain the corresponding parent vertex as a singleton piece.

The pieces are nonempty paths. They are pairwise disjoint: different Boolean blocks are disjoint by Theorem 6, and different child subtrees are disjoint because the parent relation is a tree. Their internal adjacencies come from the parent Gray cycle or the inductive child cycles, and consecutive pieces are joined by the two cross edges in the assigned diamond. The cyclic last-to-first adjacency is inherited from the parent Gray cycle. Hence flattening the cyclic list of pieces gives a single cycle.

Every vertex of the block of Q occurs exactly once, and every child subtree occurs exactly once, so the support is the whole subtree. By Lemma 10, no child uses the outgoing occurrence of Q; that edge therefore remains available. This establishes the inductive invariant. □

Theorem 12 (Uniform even-order construction)**.** *For every even $n \geqslant 4$, $\mathrm{NC}_R(n)$ has a Hamilton cycle.*

Proof. Apply Lemma 11 to the empty matching. By Lemma 8, every canonical matching lies in its subtree. By Theorem 6, the corresponding Boolean blocks partition all of $\mathrm{NC}(n)$. The assembled root cycle is therefore spanning. Since $n \geqslant 4$, $|\mathrm{NC}(n)| = \mathrm{Cat}_n \geqslant 3$, so the closed spanning walk is a Hamilton cycle in the simple graph. □

The proof is uniform at $n = 4$ and $n = 6$; no exhaustive computation is used for either order.

6 Proof of the classification

Proof of Theorem 1. For even $n \geqslant 4$, apply Theorem 12. For odd $n \geqslant 3$, apply Theorem 4.

It remains to inspect $n < 4$. There is one noncrossing partition for $n = 0$ and one for $n = 1$; the formal closed-spanning-walk convention regards each singleton graph as Hamiltonian. For $n = 2$, the refinement graph consists of one edge on two vertices and has no simple Hamilton cycle. For $n = 3$, the odd obstruction applies (equivalently, the five-vertex bipartition is unbalanced). These cases give exactly the stated disjunction. $\square$

7 Machine-checked correspondence

The accompanying Lean 4 package follows the proof above rather than the chronology of its discovery. Table 1 lists the exact declarations consumed by each paper statement. The file `paper/theorem-map.json` is the machine-readable source of this table, and the package checker verifies that every paper label and every declaration is present.

Table 1: Paper statements and Lean declarations.

Paper label	Lean declarations
`lem:bipartite`	`NCRefinementGraph_isBipartite`
`thm:signed-count`	`signedDyckSumTree_closed_form`, `signedNCCount_univ_eq_signedDyckSumTree`
`thm:odd-obstruction`	`NCRefinementGraph_fin_odd_geq3_not_isHamiltonian`
`lem:key-fibers`	`commonKey_root_or_chord`, `point_in_chord_hull`, `point_avoids_nested_chord`, `root_point_avoids_chord`
`thm:boolean-decomposition`	`decodeState_bijective`, `stateEquivNC`, `decodedNC_mergesTo_insert`
`lem:block-gray-cycle`	`rankGrayCycle`, `rankGrayVertices_nodup`, `rankGrayVertices_chain`
`lem:canonical-block-tree`	`parent_card`, `parentOption_eq_none_iff`, `inSubtree_root`
`lem:joining-diamond`	`joiningDiamond_nonempty`, `joiningDiamond`
`lem:port-injection`	`portAssignedToPair_exists`, `assignedPair_injective`, `outgoingPair_not_assigned`
`lem:tree-assembly`	`EdgePieceSystem.toCycleCode`, `subtreeCycle`, `rootCycleCode_support`
`thm:uniform-even`	`NCRefinementGraph_fin_even_geq4_isHamiltonian_petersen`
`thm:classification`	`NCRefinementGraph_fin_isHamiltonian_iff`

The publication root is `Hamilton.HamiltonianCycles`. The command `lake env lean Hamilton/HamiltonianCyclesAxiomAudit.lean` reports for the uniform even theorem, the odd obstruction, and the complete classification exactly

```
[propext, Classical.choice, Quot.sound].
```

These are standard Lean/Mathlib logical dependencies. No project axiom, `sorry`, `admit`, or `native_decide` occurs in the publication dependency closure.

The immutable technical archive corresponding to version v1.0.0 of the manuscript and formalization is available at doi:10.5281/zenodo.21316751.

8 Concluding remarks

The obstruction and construction meet exactly. Odd orders fail for the global reason forced by the bipartition sizes. Even orders admit a cycle because every Petersen block has odd Boolean dimension and because the lexicographic chord tree admits conflict-free explicit joining ports. The central point is that the proof never asks for a Hamilton path with arbitrary prescribed endpoints inside an opaque fiber. Instead, its endpoints are fixed Gray edges, and the cross-block edges are constructed together with those ports in a literal refinement diamond.

References

[1] E. Hartung, H. P. Hoang, T. Mütze, and A. Williams, Combinatorial generation via permutation languages. I. Fundamentals, *Trans. Amer. Math. Soc.* **375** (2022), 2255–2291; arXiv:1906.06069.

[2] C. Huemer, F. Hurtado, M. Noy, and E. Omaña-Pulido, Gray codes for non-crossing partitions and dissections of a convex polygon, *Discrete Appl. Math.* **157** (2009), 1509–1520.

[3] H. P. Hoang and T. Mütze, Combinatorial generation via permutation languages. II. Lattice congruences, *Israel J. Math.* **244** (2021), 359–417; arXiv:1911.12078.

[4] G. Kreweras, Sur les partitions non croisées d'un cycle, *Discrete Math.* **1** (1972), 333–350.

[5] A. Merino, Namrata, and A. Williams, On the hardness of Gray code problems for combinatorial objects, preprint, 2024; arXiv:2401.14963.

[6] T. Mütze, Combinatorial Gray codes—an updated survey, *Electron. J. Combin.* **DS26** (2024); arXiv:2202.01280.

[7] T. K. Petersen, On the shard intersection order of a Coxeter group, *SIAM J. Discrete Math.* **27** (2013), 1880–1912; arXiv:1108.5761.

[8] R. Simion, Noncrossing partitions, *Discrete Math.* **217** (2000), 367–409.

[9] R. Simion and D. Ullman, On the structure of the lattice of noncrossing partitions, *Discrete Math.* **98** (1991), 193–206.